

any further good physics e.g. nuclear material is very dense B1 [2]

7 (a) (i) $P = Vi$ C1
 $1200 = 240 \times i$ M1
 $i = 5.0 \text{ A}$ A0

(ii) $V = iR$
 $240 = 5.0 \times R$ C1
 $R = 48\Omega$ A1 [4]

(b) (i) p.d. = $(5.0 \times 4.0 =) 20 \text{ V}$ A1

(ii) mains voltage = $(240 + 20 =) 260 \text{ V}$ A1

(iii) $P = (20 \times 5.0 =) 100 \text{ W}$ A1 [3]

(c) power input = $1200 + 100 = 1300 \text{ W}$ C1
efficiency = $1200/1300 = 0.92$ A1 [2]